Analysis of the Effects of Housing Permits on Population Growth from 2010-2022: Gradient Boosting and SHAP Models

Introduction

Population growth is one of the most important factors reflecting how communities change over time, as it can affect (and be affected by) labor market trends, infrastructure quality, public services, and demand for housing, making it crucial to understand the factors that have the strongest effects on population growth. One of the most well studied variables is the availability of housing, as the availability of housing can not only affect the price of housing (thereby affecting the cost of living) but also the physical capacity for a city to hold more people and hire more people within the area itself. House permits, which are a very strong correlate of housing construction, give a clear indication of changes in housing supply, and can be used to examine the relationship between construction and population growth. Additionally, house permitting is affected by a multitude of factors, ranging from the amount of land left for zoning, to the regulatory environment in the area. However, population growth is affected by many socioeconomic, demographic, and labor market factors, making the relationship between permitting and population growth far too complicated for simple statistical modeling.

The goal of this project is to evaluate the relationship between house permitting (a strong correlate of house construction) and population growth across U.S. counties from 2010-2022. Using county-level housing, socioeconomic, demographic, and labor market data, this project applies multiple statistical approaches, and machine learning methods to determine the relationship from different perspectives, including linear relationships, nonlinear prediction methods, growth forecasting, and causal inference. This specific report uses gradient boosting

techniques to determine whether nonlinear relationships between permits, other county-level variables and population growth improve the predictive performance of different models. Subsequently both feature importance and SHAP analysis are used to find the variables which are the most important for predicting population growth, with SHAP identifying the direction and magnitude of each variable's effect on population growth.

Data

In order to determine the effect of housing permits on population growth, there were many covariates that were used for different reasons. These included the total permit rate per 1,000 people (obtained from the total number of permits in the county, divided by the population, multiplied by 1,000), as well as the number of single family permits per 1,000, the number of multifamily permits per 1,000 (specifically 2-4 unit permits), and the number of permits for buildings with at least 5 units (per 1,000). Each permit-rate variable was used as the main independent variable for its own regression, while the total permit rate was used for every other type of model. Additionally, the remaining housing variables included county-level homeownership rates, the growth rate in occupied housing units (which was lagged by a year), rent burden (the percentage of people paying more than 30% of their income in rent), and median house value. The labor market and economic characteristics that were used included median household income and the employment population ratio. The demographic variables that were used were the percent of people in a county who are white, the percent of people in a county who are black, the percent of people born outside the US, the change in the percent of people born outside the US (essentially functioning as net migration), as well as the county's median age. Lastly, I also used population density for the counties, which functions as a ceiling on how many

people can live in an area before geographic constraints inhibit housing construction (albeit, for many counties, that limit hasn't been reached). After finding the correct specifications for all of the variables, I also added two interaction terms, permits multiplied by housing growth, and permits multiplied by median house value.

Data Preparation

With the exception of the permit rate, I gathered all of the data from the American Community Survey's 5 year estimates. I used 5 year estimates instead of 1 year estimates because ACS data becomes very noisy when applied to small counties due to their small sample sizes for data collection. For the permit rate, I gathered information on total permits from the Department of Housing and Urban Development, which has the data from 1980-2022, after which I divided the number of each permit type by the population, and I multiplied them by 1,000 to get the per 1,000 rate. I used data from 2010-2022 for almost all of the variables because much of the ACS data starts in 2010, while the housing permit data set runs up to 2022. I also used 2023 data for population, foreign-born population, and the number of occupied housing units. These were used to create growth rates which were mostly used for lags so that I could use the 2022 data to determine trends. The data for the variables (from ACS data) was collected using API keys, and they were merged using SQL joins (mostly inner joins), with some variable transformations and calculations being performed in SQL using window functions, with other transformations being performed either in Excel or in Python workbooks.

For all of the models, I had to make some decisions about what data to include and what to remove. Firstly, since the COVID-19 pandemic falls within the time frame of this project, there are some years where population growth is significantly lower as a result of a crisis that

can’t easily be seen in the data. The period when this issue was most apparent was 2020-2021, as population growth was significantly lowered from 2020 to 2021 as hundreds of thousands died from the virus. While even more people died in 2021, this ended up having less of a deleterious effect on population growth because of the substantial increases in immigration during 2021. Since corresponding population growth was the variable of choice, population growth from 2020 to 2021 was associated with the values for each of the variables in 2020. As a result of the issues that 2020 would present for the data, I made the decision to remove the 2020 data from consideration, as well as its corresponding population growth. Although this made models like forecasting far more difficult, the alternative was worse as a whole year’s worth of data would be of incredibly poor quality.

Additionally, since there were some very noticeable outliers which could heavily alter the relationship between an independent and dependent variable, I tended towards removing them. This was firstly done by removing counties with a population below 2,500 as population growth figures are incredibly noisy at such low levels, since even a couple families deciding to move out can have a substantial effect on population growth. Additional outliers which were removed included counties which allowed more than 20 permits per 1,000 (a sample of roughly 30 observations), counties which had population growth rates above 8%, and below -5%, (a sample of roughly 40 observations), and counties which saw their number of available housing units either increase by more than 15% in one year, or decrease by more than 12.5% in one year (a sample of roughly 160 observations). While these removed a meaningful amount of data (roughly 0.6% of the data in the sample), even small amounts of outliers were capable of changing the best relationship between an IV and a DV from a linear relationship to a

logarithmic one, which is especially problematic for something like regression modeling, where the wrong relationship being specified can make models perform far worse.

Variables

Besides the house permit rate, which is the main independent variable for the project, the housing variables in question were used to communicate how affordable housing is in some way, or to communicate the availability of housing in full. Median house value, rent burden, and homeownership rate are used for different reasons to tell how affordable housing is in terms of the price of housing, the household level constraints from owning housing, or how many people own homes (because many people who don't own houses would own them if they could). The growth in housing units is a reflection of the housing stock at any given time, and even though it could be highly correlated with permitting frequency, it isn't, even when it's lagged.

The employment population ratio was used because it's a simple reflection of how many people work and is superior to the employment rate in some respects because it can better account for an area having lots of discouraged workers, retirees, or limited employment opportunities. Median household income was used to determine how wealthy the average person is in an area.

When it came to demographic variables, I used the three groups (white people, black people, and foreign born people) for different reasons. There is a well documented history of white people moving out of urban areas and into suburban areas when they become more racially diverse, especially with respect to black people moving into an area, and this pattern has been present as suburbs have become more diverse as well (even in middle class neighborhoods), so a higher share of black people in an area may mean that white people move to a nearby

county[1][2][3]. I used the white percentage to convey the level of racial homogeneity which may be a correlate of increased housing values for the same reason that an increased presence of minorities may be a correlate of decreased house values. I used the foreign born share not only as a variable related to diversity, but also as a way to show a crude form of net migration which is itself a massive component of population growth that may not be captured by the data in any other way, especially relevant because the reason for immigrants coming to a certain county aren't likely to be properly predicted from the data. While immigrant flows are somewhat correlated with local labor market conditions, the nature of a labor market can only partially be inferred from the data in question[4]. Lastly, I used population density to communicate the extent to which a county has already been urbanized, its ability to get more permits approved, and its capacity to become more densely populated. I chose this as a measure instead of an urban vs rural percentage because the definition of an urban area has become more restrictive with time, including through an increase in the minimum population requirement.

Methodology

In order to make the proper models, it's critical to ensure that each variable is correctly specified. This is done by checking for the shape of each variable when plotted against population growth, along with the R^2 and the RMSE (root mean squared error) having a minor role in determining whether a transformation is best. However, if the relationship between a given variable and population growth looks linear, I defaulted to using a linear transformation to avoid the risk of fixating too much on small deviations in R^2 and RMSE. Since population growth is more or less normally distributed, it does not require a logarithmic transformation. Therefore, the other variables are measured against a dependent variable which has a linear

specification. Ultimately, for most variables, a linear specification was adequate. This includes permits per 1,000 (which applies to all 4 of the permit variables), homeownership rate, the employment population ratio, the change in occupied housing units, the change in the percent of foreign born people in a county, the rent burden, the median house value, and the median age in years, as well as the two interaction terms which used linear specifications for both of their variables. Logarithmic specifications were used for the foreign born population share, as well as population density. A quadratic specification was used for the percent of people in a county who are white, and a cubic specification was used for the percent of people in a county who are black, while a reciprocal specification was used for the median household income.

Gradient boosting is an ensemble machine learning method that continuously trains models to mitigate prior mistakes by giving increased weights to misclassified instances, thereby forcing the model to increasingly examine prior mistakes in order to make changes, improving model performance[5]. It combines weak prediction models (decision trees) and sequentially trains the ensemble on those weak models in order to minimize errors and improve predictive performance. As a result of combining numerous decision trees (be it in classification or regression models), gradient boosting is highly capable of understanding and expressing nonlinear relationships between variables (often called features). Gradient boosting models are focused on minimizing loss functions, which measure the difference between model predictions and actual values[6]. However, this can make them vulnerable to overfitting, which is when a model focuses too much on replicating the patterns from the data it's trained on, thereby failing to understand patterns shown in data, leading to the model failing to generalize to new data. In order to mitigate this problem, a critical component of gradient boosting is hyperparameter

tuning wherein the gradient boosting model tests out different combinations of parameters to see which one will perform the best in terms of diagnostic tests like R^2, RMSE, or MAE.

For my model, the parameters that I used were the max_depth, learning_rate, n_estimators, subsample, colsample_bytree, min_child_weight, gamma, reg_alpha, and reg_lambda. Max depth refers to the depth of a tree, with higher values corresponding to trees that go down more levels, but run the risk of overfitting, while trees with lower values can underfit and may not capture all of the relationships between variables[7]. Learning rate refers to how much each new tree contributes to the model, with smaller values creating models that are very robust and don't easily adapt to small amounts of new information, but require long training periods, while high values speed up training but may overvalue newer data, thereby risking overfitting. The number of estimators refers to the total number of trees used in the gradient boosting model. The subsample refers to the fraction of data that each tree uses, while colsample_bytree (column sample) refers to the fraction of variables that each tree uses (which is used to force the model to adapt constraints in the amount of data that it's given).

Minimum child weight is the minimum sum of instance weights needed for a child[8]. If a decision tree partition creates a node where the total instance weight is less than min_child_weight, then the partitioning ends. In linear regressions, this refers to the minimum number of instances for each node in a decision tree. As min_child_weight increases, the gradient boosting algorithm will be more conservative. Gamma refers to the minimum reduction in losses for a tree split to occur, with higher values creating trees that only split when significant losses occur, functioning as a regularization check. Reg_alpha (also known as L1 or lasso) functions as regularization for leaf weights, helping to make a more sparse model by making some weights zero, with higher values of alpha increasing the strength of regularization.

Reg_lambda (also known as L2 or ridge) functions as regularization for leaf weights as well, smoothing out weights without making a more sparse model, with higher values leading to models that are more resistant to noise.

Optuna was used to search for the optimal values across each of the parameters. While optuna will find a model that performs very well, it will also find slightly different parameters each time you run the model, even after implementing a random state, where the exact same data is put in the training group every time. As such, it's important to also judge the model based on its RMSE. After the parameters were found, I used feature importance to determine which variables had the strongest contributions to population growth. Additionally, the same logic was used to make group shuffle splits, and year shuffle splits. Group shuffle splits essentially randomly assign certain counties to the training group, and the testing group, at which point all of their yearly values will either be in one group or another. Additionally, year shuffle splits were used, assigning all of the values of a certain year in the train or test group (in the case of this model, the years put in the test group were 2014, 2016, and 2018).

These two changes were in contrast to the baseline model, which randomly assigns all counties to the train or test groups, meaning some of the values in a county will be in the training set, and some in the testing set, and this is random for each county. The two alternative models I used were to see if the pattern changed when you only trained on certain years, or certain counties, in order to see if the model was better at handling large numbers of missing values which aren't affected as much by the information which is given to the training set. Essentially, while taking one random year away from a county means the model will be able to fill in the blanks pretty easily, taking a specific year away from all counties means the model has to perform more work to understand how the relationship between all the variables changed in a

given year in the absence of any information, and the same is the case with the model having to fill in the blanks and approximate the relationships for whole missing counties.

However, while feature importance lets you know which variables are most important, it doesn't give you magnitudes on how much each variable matters, nor does it tell you the direction in which the relationship goes. As a result, SHAP (SHapley Additive exPlanations) was used to provide interpretable values measuring the impact of the inputs (independent variable) on population growth[9]. SHAP is used to pair the interpretability of a regression coefficient with the feature importance that a XGBoost model can help provide. While knowing the regression coefficients in a standard linear regression can help, they only tell you so much. They don't explicitly tell you how much y changes when you increase an independent variable by one standard deviation, or how much variance a certain independent variable has, which is what makes descriptive statistics so important to pair alongside regression models. Meanwhile, SHAP values are used to tell you how much the y value of an observation changed as a result of the difference between the baseline value of an independent variable, and the actual value of that variable for an observation. What it does is essentially tell you the numerical effect of median house prices being twice as expensive as the baseline county, and not just more expensive. While SHAP values can be used to make waterfall plots which show the effect of the value for each independent variable on the predicted value of the dependent variable, SHAP has its own importance metric, similar to feature importance used after the completion of a gradient boosting model. After using SHAP, I displayed partial dependence plots for the 3 variables with the highest SHAP importance values (essentially the 3 variables which had the largest effect on population growth). Partial dependence plots show you a graph of how changing the values of a variable affect the predicted value of the output.

Results

The optimal parameters from the gradient boosting model were a max depth of 8, a learning rate of roughly 0.00642 1,808 trees used, a subsample of roughly 0.51, a column sample of 0.70, a minimum child weight of 2, a gamma of 0.253, a reg_alpha (L1 regularization) of 0.281, a reg_lambda (L2 regularization) of 2.66, and the best RMSE was 0.67 (for the test data, the RMSE was actually lower). For feature importance, the most important variables were the permit house value interaction term, the permit housing unit growth interaction variable, the total permits per 1000, median house value, and median age, which was the same order for the group shuffle split and the yearly split as well. The tables below show the feature importance and SHAP importance values for the three models, and the diagnostic results for the gradient boosting models:

Baseline Model: Feature Importance	Group Shuffled Model: Feature Importance	Yearly Shuffled Model: Feature Importance
Permits*housing_value: 0.277	Permits*housing_value: 0.275	Permits*housing_value: 0.261
Permits*growth_in_housing_units: 0.118	Permits*growth_in_housing_units: 0.122	Permits*growth_in_housing_units: 0.125
Permits per 1000: 0.0825	Permits per 1000: 0.0885	Permits per 1000: 0.0768
Median House Value: 0.0577	Median House Value: 0.0599	Median House Value: 0.0553
Median Age: 0.0530	Median Age: 0.0524	Median Age: 0.0552
Model R^2: 0.664	Model R^2: 0.507	Model R^2: 0.652
Test RMSE: 0.586	Test RMSE: 0.708	Test RMSE: 0.585

Figure 1: Feature Importance Values and Diagnostic Test Results for the Baseline Gradient Boosting Model, as well as the Models Based on Group Shuffle Splits, and Yearly Shuffles.

Baseline Model: SHAP Importance	Group Shuffled Model: SHAP Importance	Yearly Shuffled Model: SHAP Importance
Permits*housing_value: 0.211	Permits*housing_value: 0.212	Permits*housing_value: 0.209
Median age: 0.170	Median age: 0.175	Median age: 0.178
Median House Value: 0.154	Median House Value: 0.155	Median House Value: 0.145
Homeownership rate: 0.102	Homeownership rate: 0.103	Homeownership rate: 0.0993
Reciprocal (Median Household Income): 0.0945	Reciprocal (Median Household Income): 0.0949	Reciprocal (Median Household Income): 0.0983

Figure 2: SHAP Importance Values for the Baseline SHAP Model, as well as the Models Based on Group Shuffle Splits, and Yearly Shuffles.

As was the case with the regression model, and the neural network model, the relationship between the R^2 of the model, and the slope of the predicted vs actual values of population growth were highly correlated. When it comes to SHAP values, counties with very low or very high values with respect to certain variables can see their population growth rates driven significantly by factors like population density, and foreign born population, as seen in the example below:

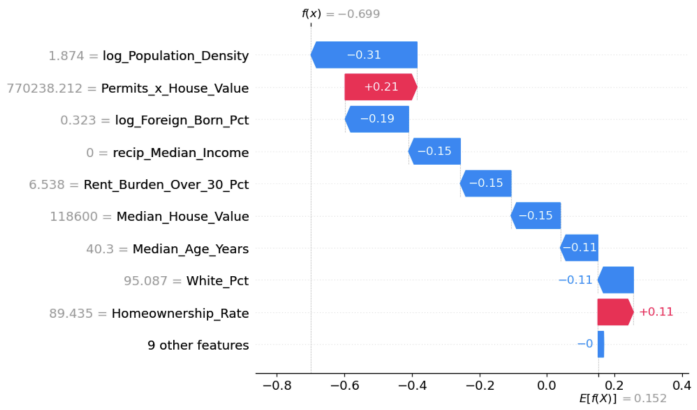


Figure 3: SHAP Waterfall Plot Based on A Sample Observation For Cameron Parish: Louisiana in 2011 With the Blue Bars Illustrating Negative Contributions to the Population Growth Rate and the Red Bar Illustrating Positive Contributions.

In the case of the SHAP contributions shown above, the county in question is more racially homogenous, has a higher homeownership rate, a lower median house value, a lower rent burden, an almost identical median age, fewer immigrants, lower population density and a higher value for the interaction variable than the average county.

Discussion

When comparing the gradient boosting models, the three models found that the same few variables were the most important in determining population growth. When evaluating the models based on diagnostic tests, the baseline model and the yearly shuffle model performed almost identically, with the group shuffle model performing noticeably worse. This isn't very surprising considering the amount of variation between the counties in this sample, as there are counties with anywhere from 5,000-5,000,000 residents. As was the case with the gradient boosting models, the SHAP models returned somewhat similar results when it came to determining the variables which have the largest effects on population growth. The interaction terms were both very important in the gradient boosting model, while only one of those terms was important in the SHAP model, and in the regression models, the terms were not very relevant.

In practice, SHAP values and feature importance values tell similar things, while also not being interchangeable. While gradient boosting does a better job at telling you what variables are most important to determining model performance (in this case referring to their ability to predict population growth), SHAP values reveal important information about the direction and magnitude of the relationship between each independent and dependent variable.

The fact that the two models reveal different information about the most important variables may be a result of the fact that SHAP values have a similar function to regressions. The aforementioned SHAP waterfall plot on Cameron Parish: Louisiana in the results section helped reveal information about how each variable affected population growth in ways that are somewhat similar to regression models, while still serving a function that regression models can't do. However, that was only one plot, and the individual effects of each variable on the outcome are highly dependent on how much the values of each variable differ from their means. Even with all that said, just because a given value for a variable has a positive SHAP contribution in one case doesn't mean it will have a positive SHAP contribution in every case, making SHAP models a little more idiosyncratic than regressions, but still not without their merits.

Ultimately, the regressions in question generally found that the interaction terms were not nearly as important as many of the other variables. In the gradient boosting portion, the permitting and median house value interaction term was by far the most important variable in each of the three models, but in the SHAP portion, it was not substantially more important than the next most important variable for any model. One of the most surprising takeaways from both types of models was the fact that the permitting rate was so much less important than it was in the regression models. In the regression portion, the permit rate had by far the highest effect on population growth when judging the effect of a one standard deviation increase, while the coefficient was also roughly 61 standard errors away from zero. However, in the gradient boosting models, it was a distant third in terms of feature importance, and in the SHAP models, it was actually one of the less important variables. A possible explanation for this is that the

permitting and house value interaction terms could be absorbing much of the impact that the variable has on population growth as was seen in the regression.

Both the SHAP models and the gradient boosting models have some relevant limitations. As was the case with the regression and neural network models, the relationship between actual and predicted population growth rates are compressed, as the slope of the relationship between the actual and predicted values of population growth is very similar to the model's R^2 value. This is a very serious issue because it means that instead of a prediction for an observation with a population growth rate of 1% returning a predicted rate of 1%, it will typically return a value considerably less than 1%. As a result, the model will tend to underpredict the growth rate in high growth counties, and vice versa. As was stated in previous discussions, this issue may be a result of the fact that population growth is a very noisy variable which is not heavily linked to a place having a high standard of living. The factors used for this project all have their importance, but so too does crime rate, public service quality, educational quality, and a slew of other variables. This project didn't compile data for those variables, and many of them are difficult to obtain at the county level, however, adding a couple variables on the subject could be very important extensions to this project.

Another limitation to these results is the fact that these models do not establish causality. Gradient boosting and SHAP are not causal models, and even if they were, there are still missing variables that could affect the interpretations of each value, and cause omitted variable bias, especially important for SHAP, which focuses more on giving precise interpretations for the way in which each variable affects the outcome.

The final models found that the variables most responsible for predicting overall population growth were the interaction terms concerning house permitting rate, along with

median house value, and the growth in available housing units. Housing permit rate was still a somewhat important variable, but its effect was far less apparent than in the regression models. These findings were relatively consistent across the data splits for both the gradient boosting, and SHAP models. This may've been a result of interaction terms capturing some of the effect that permitting rate tends to have on population growth. That being said, the policy takeaways from this model do not substantially differ from other methods, as they don't point to house permitting not being an important variable. Instead, they suggest that the predictive power of permitting may depend on how it interacts with other housing characteristics.

Appendix

The housing permits dataset was obtained from the department of housing and urban development[10]. The ACS data included the datasets for Selected Social Characteristics, Selected Housing Characteristics, Income in the Past 12 Months, Place of Birth by Poverty Status in the Past 12 Months in the United States, Age and Sex, Hispanic or Latino Origin by Race, Selected Economic Characteristics, and the Annual Geographic Information Table (GEOINFO)[11][12][13][14][15][16][17][18]. The code was based on a hyperparameter tuning with optuna video with an accompanying article with code from Kaggle[19].

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